IT 311 – Advanced Database Systems

Laboratory Exercise No. 3: Normalization and Outlier Detection

Name: Justin Von T. Vergara Frederick S. Madayag	**Score:** / 80 points
Course/Year/Section: BSIT 3A- NS	**Date:** September 8, 2025

Another important step of data cleansing is to identify unusual cases and remove them from the data set. In some situations, the outliers themselves might be the most interesting cases (detecting fraudulent credit card transactions, for example), but in most cases outliers are simply the result of an incorrect measurement and should be removed from the data set.

TASK 1: *Prepare the data (20 points)*

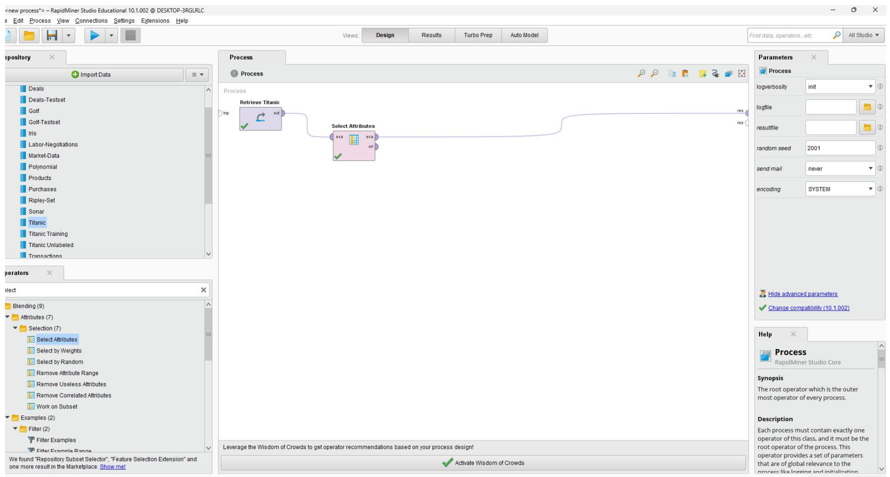
1. Drag the Titanic data into the process.
2. Add the operator ***Select Attribute*** and connect it.
3. Change the *Parameters* and remove Cabin, Lifeboat, Name, and Ticket number. Change the ***attribute filter type*** into ***a subset***.
4. In the ***select subset***, remove the four mentioned attributes above.

DID YOU KNOW?

The result will be a data set only containing those columns we believe will contribute well to our outlier detection. We will use a distance-based outlier detection algorithm which calculates the ***Euclidean distance between the data points*** and marks those points which are farthest away from other data points as outliers. The Euclidean distance uses the distances between two data points for each individual attribute. Think about it: what is the effect on the distance if the attributes have different value ranges (one attribute between 0 and 5 and another attribute between 1 and 1000)? Attributes with larger values will contribute much more than those with smaller values. For this reason, we should ensure that all attributes are using similar value ranges. This transformation is called ***Normalization.***

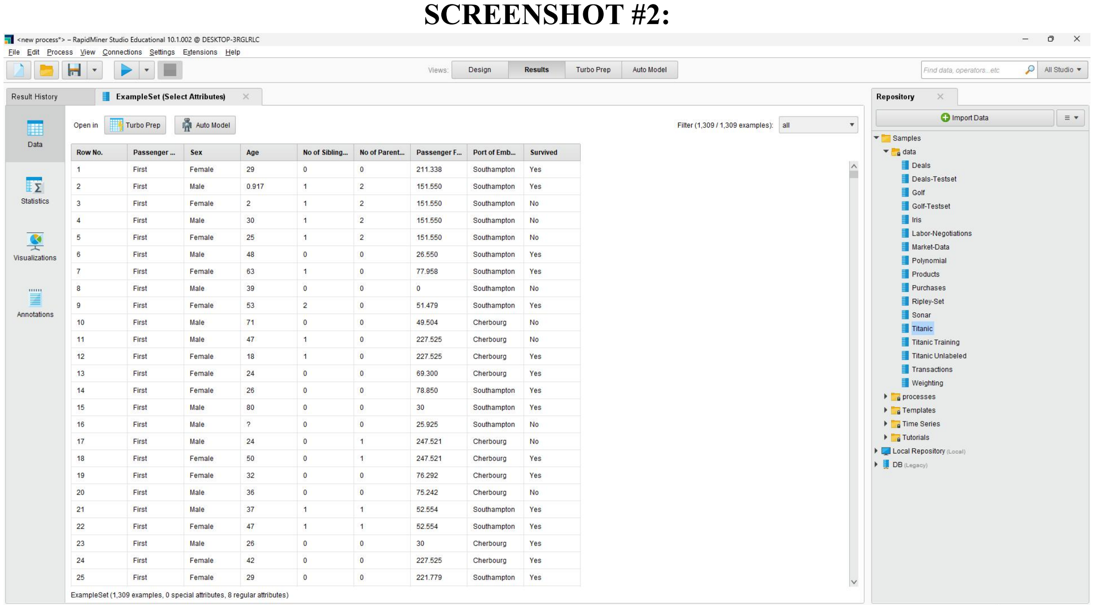
5. Take a screenshot of the process window and paste it here.

SCREENSHOT #1:



6. Run the process and take a screenshot of the remaining attributes.

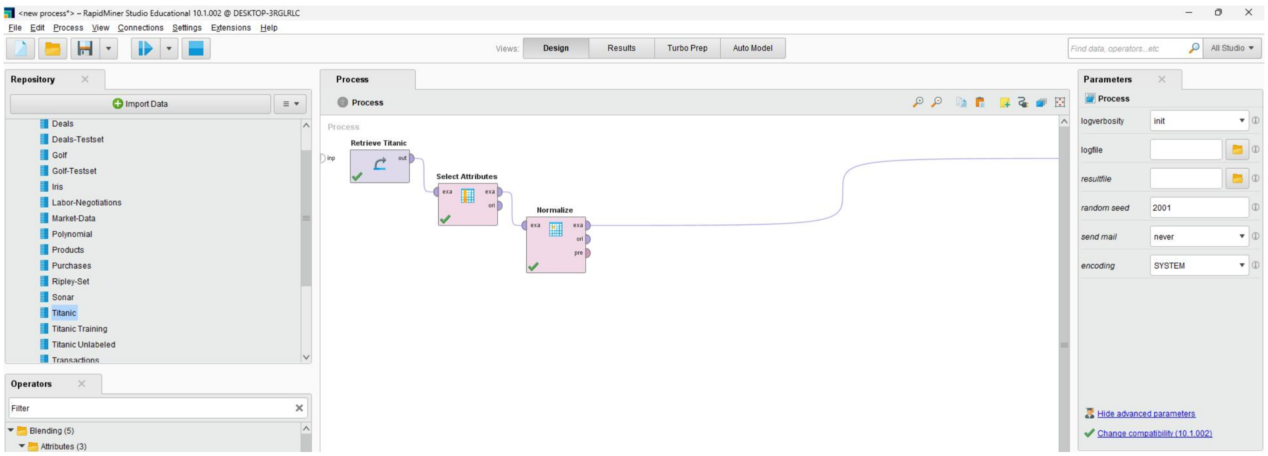
SCREENSHOT #2:



TASK 2: *Normalized the Attribute Value Ranges (10points)*

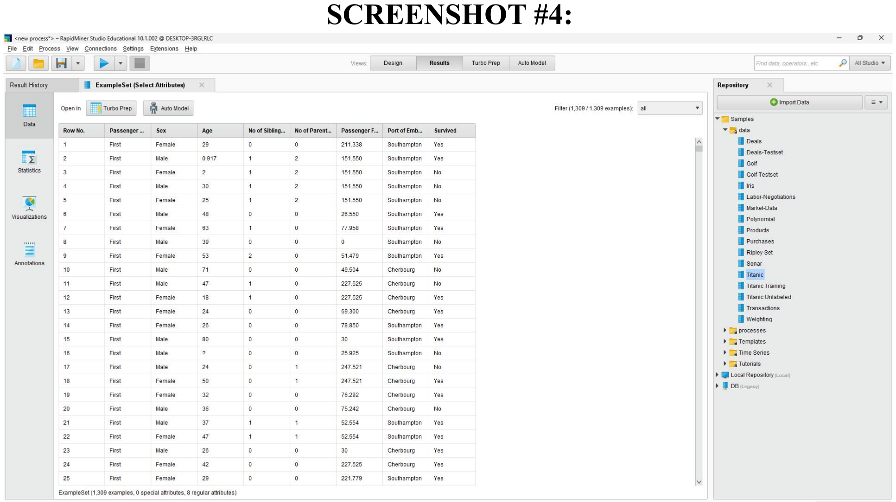
1. Add the operator ***Normalize*** and connect it.
2. Take a screenshot of the process window.

SCREENSHOT #3:



3. Run the process and take a screenshot of the result.

SCREENSHOT #4:

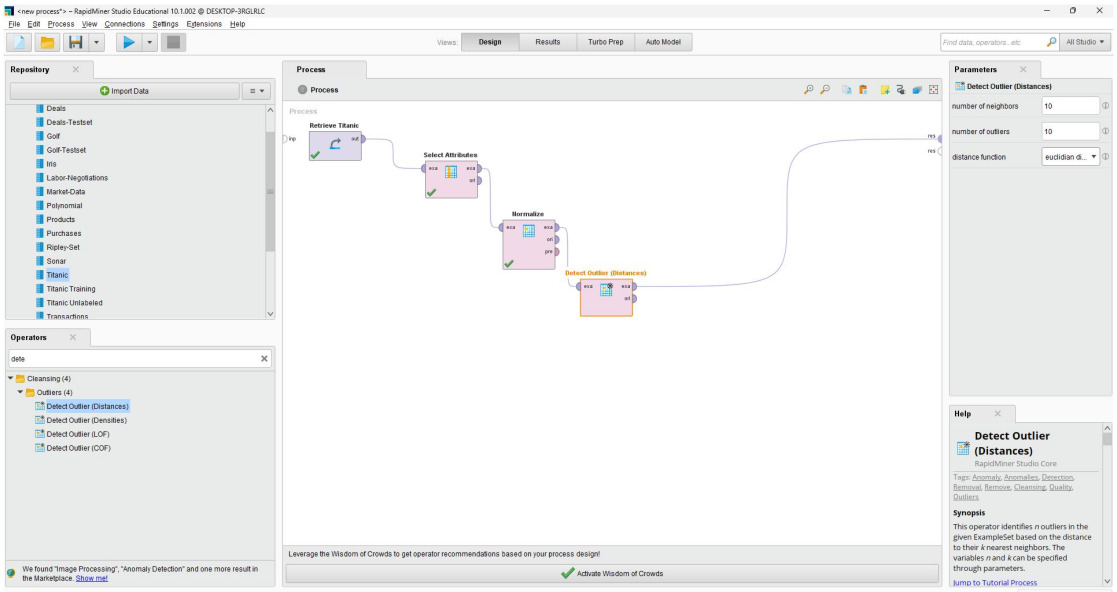


In general, you should *always* normalize your data before you apply

TASK 3: *Detect Outliers (10points)*

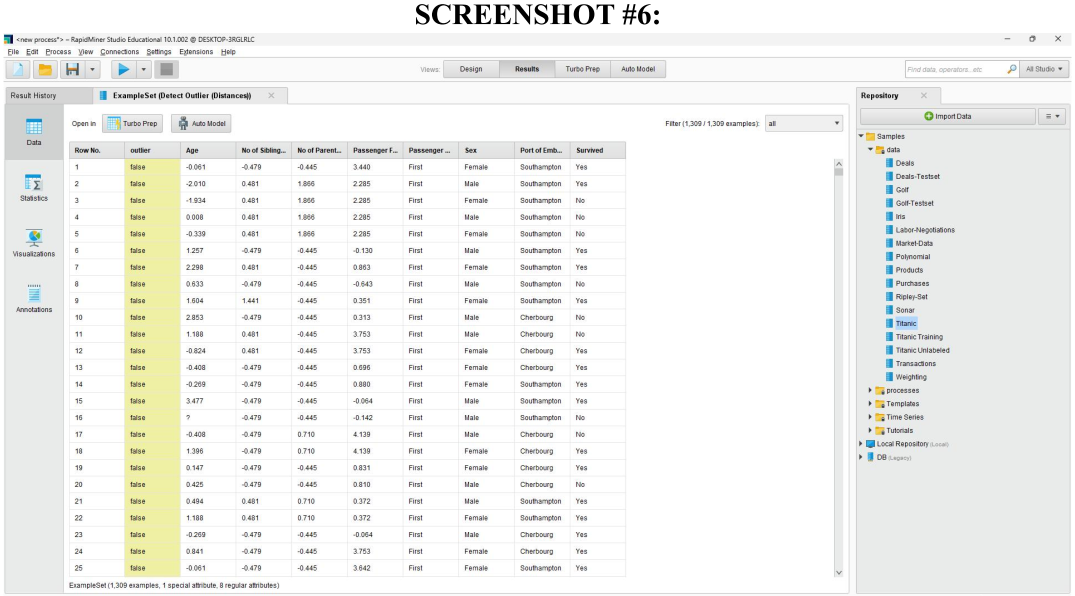
1. Search for the operator ***Detect Outlier (Distances),*** add it, and connect it to ***Normalize.***
2. Take a screenshot of the process window.

SCREENSHOT #5:



3. ***Run*** the process and take a screenshot of the result.

SCREENSHOT #6:



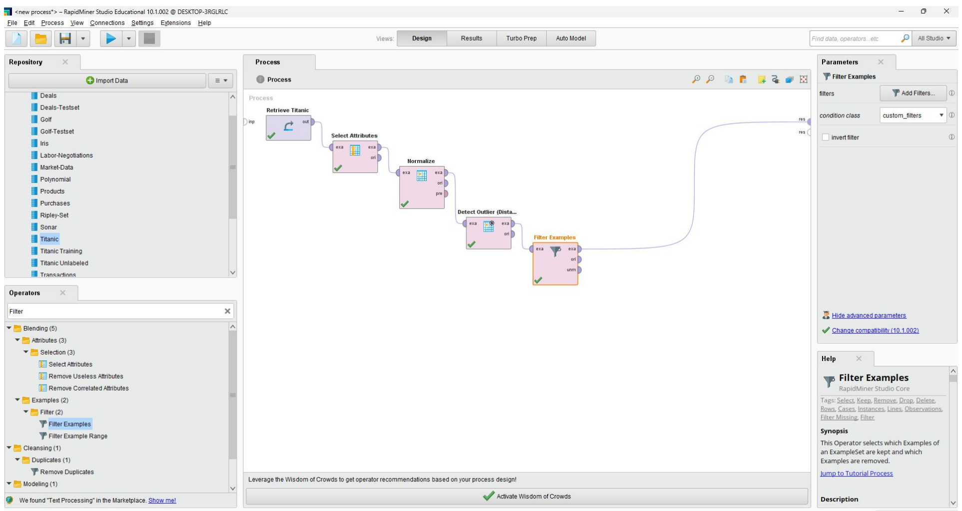
DID YOU KNOW?

The Detect Outlier (Distances)

TASK 4: *Remove Outliers from the example set. (10points)*

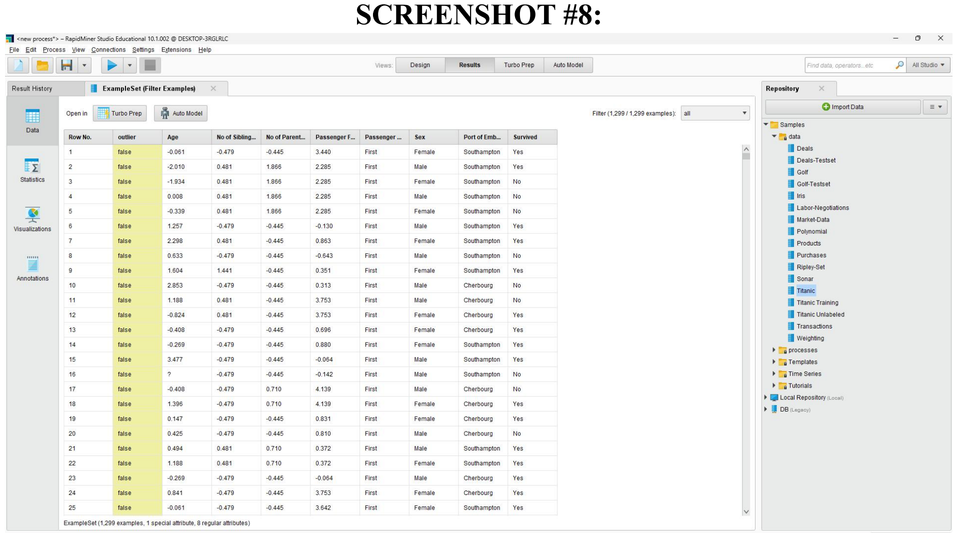
1. Add ***Filter Examples*** to the process and connect it to the previous operator and the result port on the right.
2. In its *Parameters*, add a new filter with ***Outlier, equals,*** and ***false*** as values.
3. Take a screenshot of the process window.

SCREENSHOT #7:



4. Run the process and take screenshot of the result.

SCREENSHOT #8:

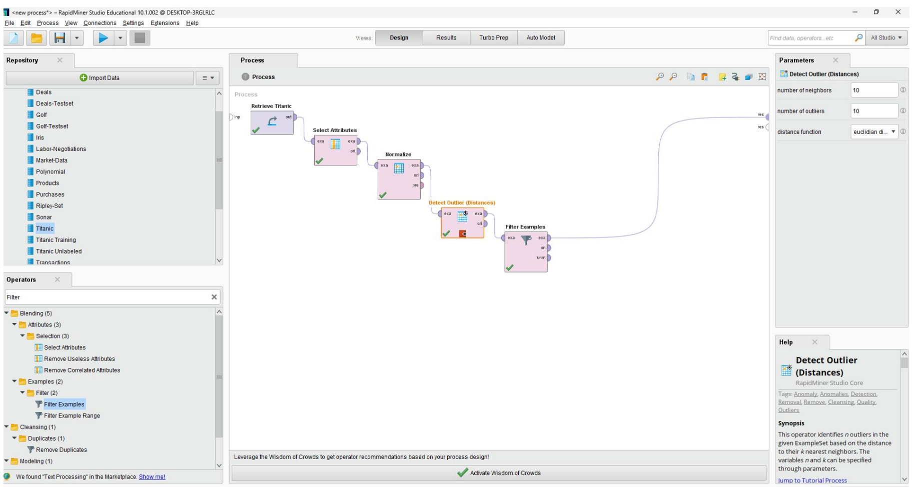


DID YOU KNOW?
When you click the run button after performing Task #4, the process

TASK 5: *Use Breakpoints to See Intermediate Results (10 points)*

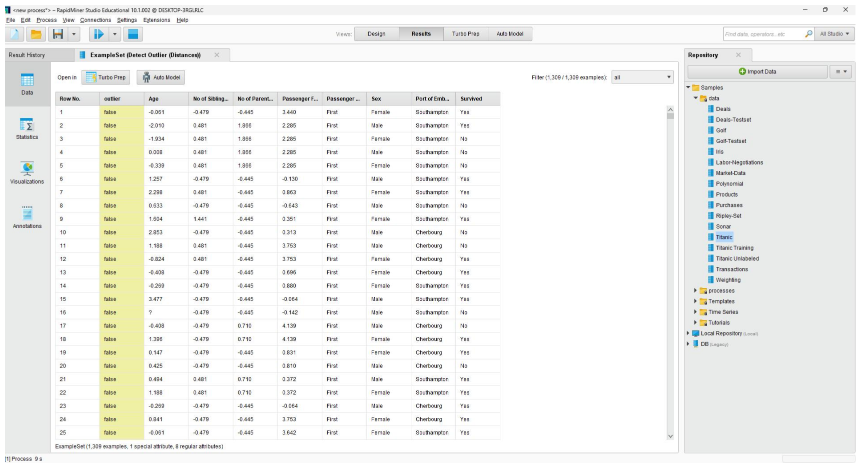
1. Go back to the design view.
2. Right click on ***Detect Outlier (Distances)*** and select ***Breakpoint after*** in the menu. Note that small icon appears at the bottom of the operator after you added the breakpoint. Take a screenshot of the process window.

SCREENSHOT #9:



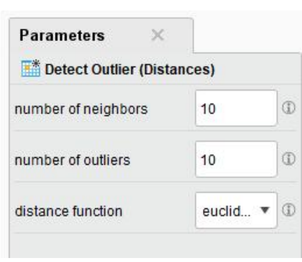
DID YOU KNOW?
The process is paused at the breakpoint and the

3. Run the process again. Take a screenshot of the process window and the result.
   SCREENSHOT #10:



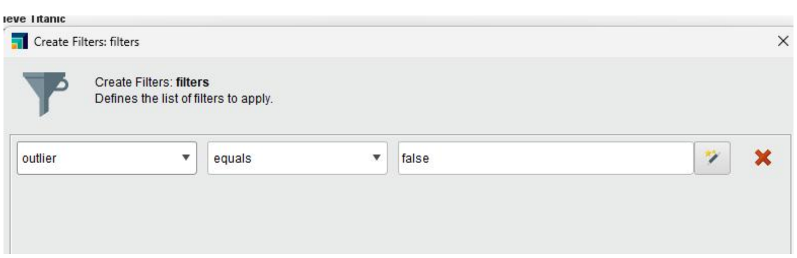
TASK 6: *Challenge (20 points)*

1. How would you change the process, so it finds 20 outliers instead of 10?
   To find 20 outliers instead3 of 10, change the parameters from 10 into 20

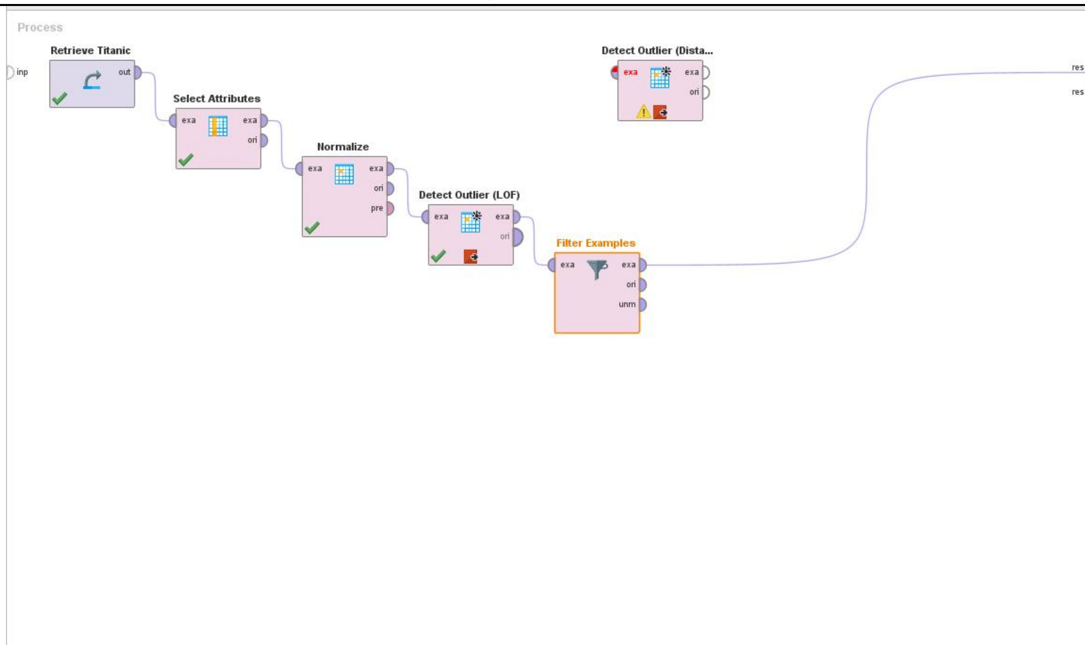


2. How can you change the process to only show outliers instead of removing them?

   To only show outliers, we just need to change the filter to TRUE instead of false.

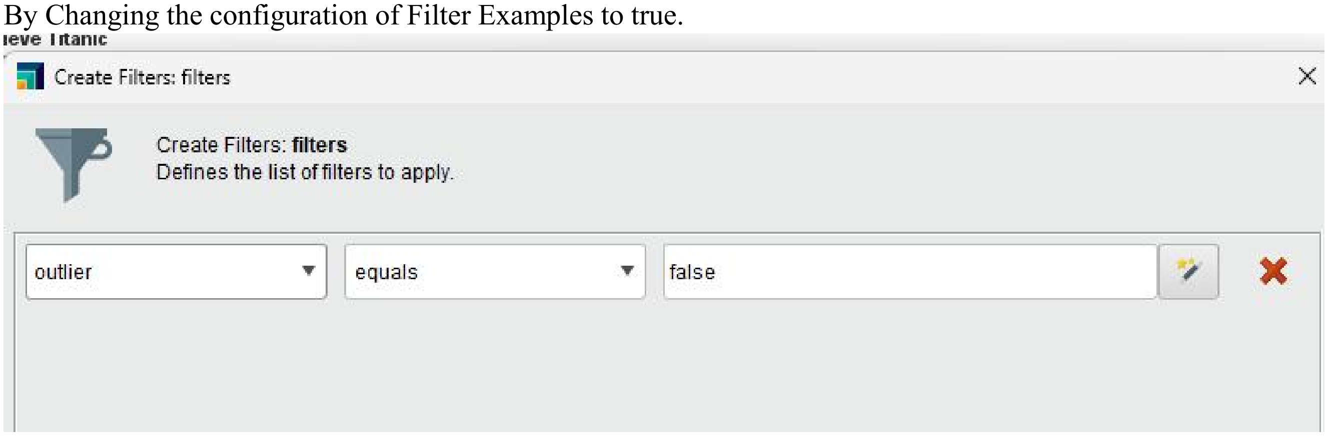


3. Replace the outlier detection operator with ***Detect Outlier (LOF)*** and a breakpoint after this operator before you execute. What is the difference to before?
   The difference is that the distance-based method marks the outliers using global distance from all points, while LOF identifies outliers based on density difference, so the detected outliers may change depending on the structure.

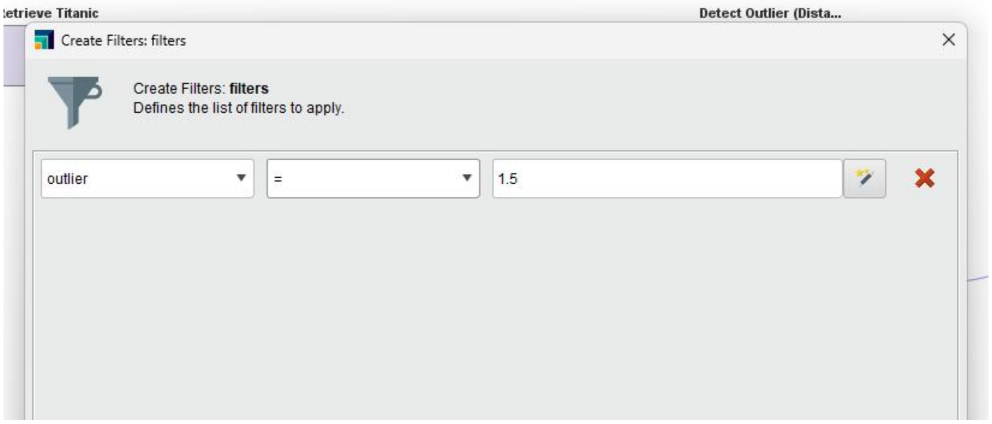


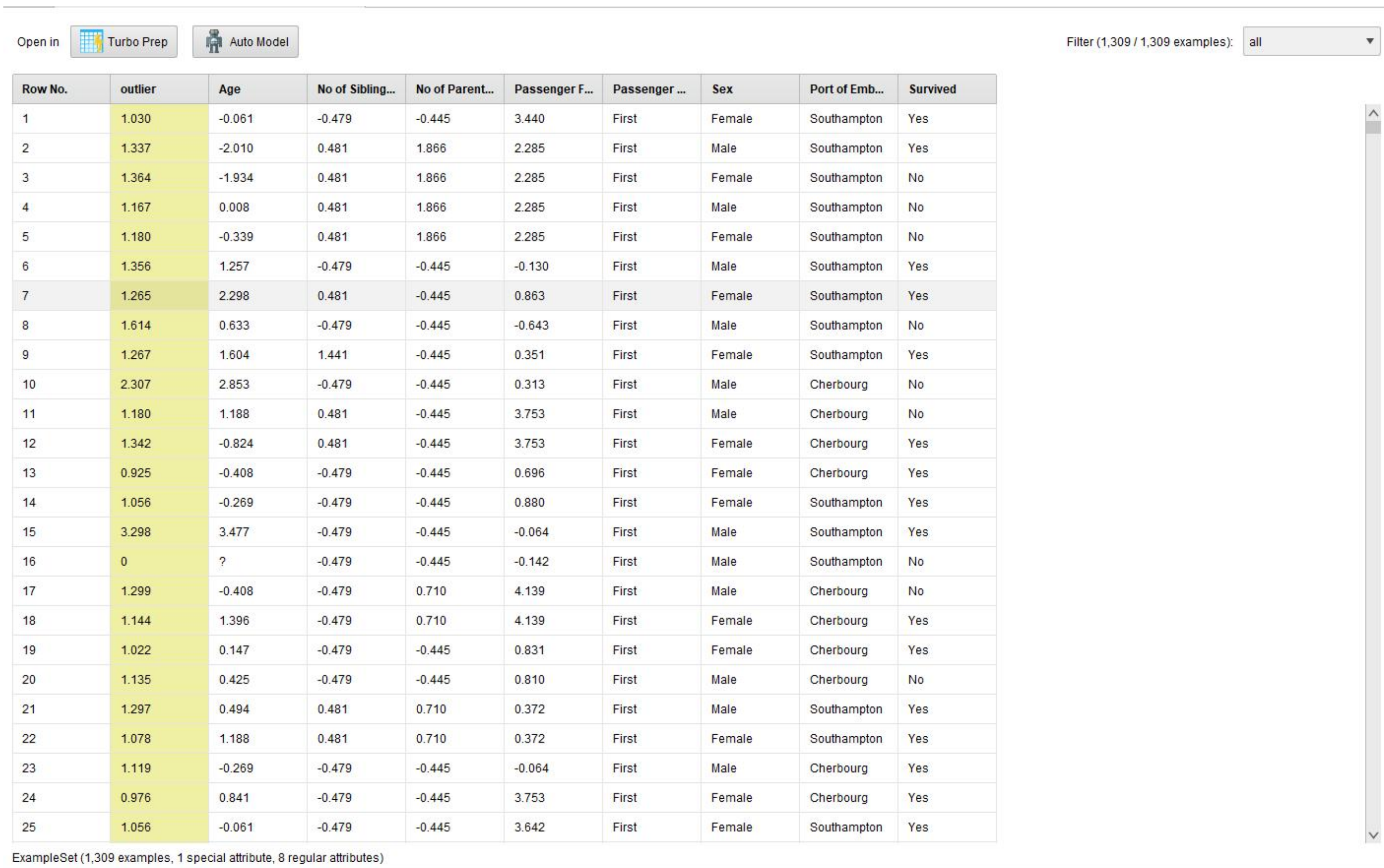
4. How do you now need to change the filter to only keep the top outliers?

By Changing the configuration of Filter Examples to true.



IF Using Detect Outlier, change the range or threshold:





Row No.	outlier	Age	No of Sibling...	No of Parent...	Passenger F...	Passenger ...	Sex	Port of Emb...	Survived
1	1.030	-0.061	-0.479	-0.445	3.440	First	Female	Southampton	Yes
2	1.337	-2.010	0.481	1.866	2.285	First	Male	Southampton	Yes
3	1.364	-1.934	0.481	1.866	2.285	First	Female	Southampton	No
4	1.167	0.008	0.481	1.866	2.285	First	Male	Southampton	No
5	1.180	-0.339	0.481	1.866	2.285	First	Female	Southampton	No
6	1.356	1.257	-0.479	-0.445	-0.130	First	Male	Southampton	Yes
7	1.265	2.298	0.481	-0.445	0.863	First	Female	Southampton	Yes
8	1.614	0.633	-0.479	-0.445	-0.643	First	Male	Southampton	No
9	1.267	1.604	1.441	-0.445	0.351	First	Female	Southampton	Yes
10	2.307	2.853	-0.479	-0.445	0.313	First	Male	Cherbourg	No
11	1.180	1.188	0.481	-0.445	3.753	First	Male	Cherbourg	No
12	1.342	-0.824	0.481	-0.445	3.753	First	Female	Cherbourg	Yes
13	0.925	-0.408	-0.479	-0.445	0.696	First	Female	Cherbourg	Yes
14	1.056	-0.269	-0.479	-0.445	0.880	First	Female	Southampton	Yes
15	3.298	3.477	-0.479	-0.445	-0.064	First	Male	Southampton	Yes
16	0	?	-0.479	-0.445	-0.142	First	Male	Southampton	No
17	1.299	-0.408	-0.479	0.710	4.139	First	Male	Cherbourg	No
18	1.144	1.396	-0.479	0.710	4.139	First	Female	Cherbourg	Yes
19	1.022	0.147	-0.479	-0.445	0.831	First	Female	Cherbourg	Yes
20	1.135	0.425	-0.479	-0.445	0.810	First	Male	Cherbourg	No
21	1.297	0.494	0.481	0.710	0.372	First	Male	Southampton	Yes
22	1.078	1.188	0.481	0.710	0.372	First	Female	Southampton	Yes
23	1.119	-0.269	-0.479	-0.445	-0.064	First	Male	Cherbourg	Yes
24	0.976	0.841	-0.479	-0.445	3.753	First	Female	Cherbourg	Yes
25	1.056	-0.061	-0.479	-0.445	3.642	First	Female	Southampton	Yes

ExampleSet (1,309 examples, 1 special attribute, 8 regular attributes)